Tutorial for Chapter 3: Part 2

Exercise 1.
A random variable X has the following PDF:

$$f(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Verify that $f(x)$ is a valid PDF.
2. Calculate the probability $P(0.3 < X < 0.7)$.
3. Write the expression for $F(x)$ for all x.
4. Calculate $F(0.5)$ and $F(1)$.
5. What is the probability $P(X \leq 0.4)$?

Solution 1.

1. First $f(x)$ is positive for $0 \leq x \leq 1$. To verify that $f(x)$ is a valid PDF, we check:
$$\int_{-\infty}^{\infty} f(x)\,dx = \int_0^1 2x\,dx = \left[x^2\right]_0^1 = 1.$$
Thus, $f(x)$ is a valid PDF.
2. To calculate $P(0.3 < X < 0.7)$:
$$P(0.3 < X < 0.7) = \int_{0.3}^{0.7} 2x\,dx = \left[x^2\right]_{0.3}^{0.7} = 0.7^2 - 0.3^2 = 0.49 - 0.09 = 0.4.$$
3. The CDF $F(x)$ is:
$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \int_0^x 2t\,dt & \text{for } 0 \leq x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$$
For $0 \leq x < 1$:
$$F(x) = \left[t^2\right]_0^x = x^2.$$
4. Thus:
$$F(0.5) = (0.5)^2 = 0.25, \quad F(1) = 1.$$
5. For $P(X \leq 0.4)$:
$$P(X \leq 0.4) = F(0.4) = (0.4)^2 = 0.16.$$

Exercise 2.
Consider the following CDF:

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{x^2}{4} & \text{for } 0 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$

1. Find the PDF $f(x)$ associated with this CDF.

2. Determine the probability $P(1 < X < 2)$.

Solution 2.

1. The PDF $f(x)$ is:

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{x}{2} & \text{for } 0 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$$

2. For $P(1 < X < 2)$:

$$P(1 < X < 2) = F(2) - F(1) = 1 - F(1) = 1 - \frac{1^2}{4} = 1 - 0.25 = 0.75.$$

Exercise 3.
The probability density of the random variable X is given by

$$f(x) = \begin{cases} \frac{c}{\sqrt{x}}, & 0 < x < 4, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the value of c;

2. Find $P(X < 1/4)$ and $P(X > 1)$ and $P(X = 2.5)$.

Solution 3.

1. According to the definition of pdf,

$$1 = \int_0^4 \frac{c}{\sqrt{x}} dx = c \cdot \frac{x^{1/2}}{1/2}\Big|_0^4 = 4c \implies c = \frac{1}{4}.$$

2. Write

$$P(X < 1/4) = \int_0^{1/4} \frac{1}{4\sqrt{x}} dx = \frac{1}{4} \cdot \frac{x^{1/2}}{1/2}\Big|_0^{1/4} = \frac{1}{4}.$$

And

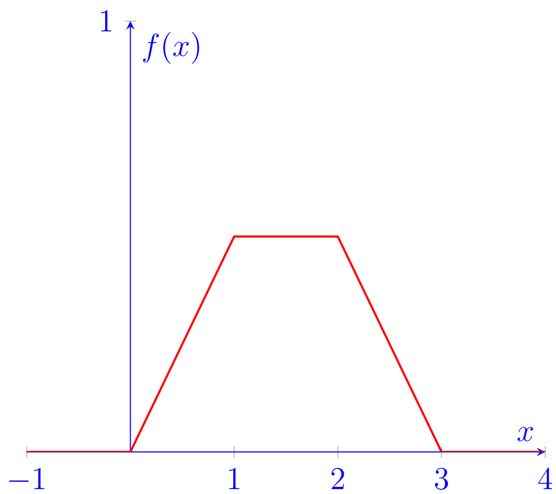
$$P(X > 1) = \int_1^4 \frac{1}{4\sqrt{x}} dx = \frac{1}{4} \cdot \frac{x^{1/2}}{1/2}\Big|_1^4 = 1 - \frac{1}{2} = \frac{1}{2}.$$

Clearly, $P(X = 2.5) = 0$ as X is continuous. $\square$

Exercise 4.
Find the distribution function of the random variable X whose probability density is given by

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x \leq 1, \\ \frac{1}{2}, & 1 < x \leq 2, \\ \frac{3-x}{2}, & 2 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Solution 4.
The graph of this pdf is



Note that, for discrete random variable $f(x_i) = F(x_i) - F(x_{i-1})$, for continuous random variable $f(x) = \frac{d}{dx}F(x)$. Thus

$$F(x) = \begin{cases} 0, & x \leq 0, \\ \frac{x^2}{4}, & 0 < x \leq 1, \\ \frac{1}{4} + \frac{x-1}{2}, & 1 < x \leq 2, \\ \frac{3}{4} + \frac{3x}{2} - \frac{x^2}{4} - \left(\frac{3 \cdot 2}{2} - \frac{2^2}{4}\right), & 2 < x < 3, \\ 1, & \text{otherwise.} \end{cases}$$

□

Exercise 5.
The loss due to a fire in a commercial building is modelled by a random variable X with density function

$$f(x) = \begin{cases} 0.005(20 - x) & \text{for } 0 < x < 20 \\ 0 & \text{elsewhere} \end{cases}$$

Given that a fire loss exceeds 8, calculate the probability that it exceeds 16.

Solution 5.

First calculate the probability that the exceeds 8

$$
\begin{aligned}
P[X > x] = \int_x^{20} 0.0005(20-t)\,dt &= 0.005\left(20t - \frac{1}{2}t^2\right)\Big|_x^{20} \\
&= 0.005\left(400 - 200 - 20x + \frac{1}{2}x^2\right) \\
&= 0.005\left(200 - 20x + \frac{1}{2}x^2\right), \quad \text{where } 0 < x < 20.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
P(X > 16 \mid X > 8) &= \frac{P(X > 16)}{P(X > 8)} \\
&= \frac{200 - 20(16) + \frac{1}{2}(16)^2}{200 - 20(8) + \frac{1}{2}8^2} \\
&= \frac{8}{72} = \frac{1}{9}.
\end{aligned}
$$

Exercise 6.

The lifetime of a machine part has a continuous distribution on the interval $(0, 40)$ with probability density function $f(x)$, where $f(x)$ is proportional to $(10+x)^{-2}$ on the interval.

Calculate the probability that the lifetime of the machine part is less than 6.

Solution 6.

We know the density has the form

$$
f(x) = \begin{cases} k\,(10+x)^{-2} & 0 < x < 40 \\ 0 & \text{elsewhere} \end{cases}
$$

First determine the proportionality constant k.

$$
\begin{aligned}
\int_0^{40} k\,(10+x)^{-2}\,dx = 1 &\implies -k\,(10+x)^{-1}\Big|_0^{40} = 1 \\
&\implies \frac{k}{10} - \frac{k}{50} = 1 \\
&\implies k = \frac{25}{2} = 12.5
\end{aligned}
$$

Then, calculate the probability over the interval $(0, 6)$

$$
\begin{aligned}
P(X < 6) = \int_0^6 f(x)\,dx &= 12.5\int_0^6 (10+x)^{-2}\,dx \\
&= -12.5\,(10+x)^{-1}\Big|_0^6 \\
&= 12.5\left(\frac{1}{10} - \frac{1}{16}\right) \\
&= 0.47
\end{aligned}
$$

Exercise 7.
An insurance company insures a large number of homes. The insured value, X, of a randomly selected home is assumed to follow a distribution with density function

$$f(x) = \begin{cases} 3x^{-4} & \text{for } x > 1 \\ 0 & \text{elsewhere} \end{cases}$$

Given that a randomly selected home is insured for at least 1.5, calculate the probability that it is insured for less than 2.

Solution 7.
The distribution function is

$$F(x) = P(X \le x) = \int_1^x 3t^{-4}\,dt = -t^{-3}\big|_1^{-3} = 1 - x^{-3}.$$

Then,

$$\begin{aligned} P(X < 2 \mid X \ge 1.5) &= \frac{P[(X < 2) \cap (X \ge 1.5)]}{P(X \ge 1.5)} \\ &= \frac{P(X < 2) - (X < 1.5)}{P(X \ge 1.5)} \\ &= \frac{F(2) - F(1.5)}{1 - F(1.5)} \\ &= \frac{(1 - 2^{-3}) - (1 - (1.5)^{-3})}{1 - (1 - (1.5)^{-3})} \\ &= \frac{37}{64} = 0.578 \end{aligned}$$